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Zhejiang University-University of Illinois Urbana-Champaign Institute

Undergraduate Thesis (Design)

Thesis Title: Image Denoising in Medical Imaging with Transformers

Student Name: Yuxing Wu

Student ID: 3210110961

Supervisor: Bo Zhao

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Content Checklist

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<tr><td>Thesis title</td><td colspan="5">Image Denoising in Medical Imaging with Transformers</td></tr>
<tr><td colspan="4">Content Checklist</td><td colspan="2">Evaluation Section (√)</td></tr>
<tr><td rowspan="12">Senior Thesis</td><td colspan="3">1. Cover (Use the unified cover, blue recommended)</td><td>Good</td><td>Fair</td></tr>
<tr><td colspan="3">2. Academic integrity Commitment of Senior Thesis</td><td></td><td></td></tr>
<tr><td colspan="3">3. Acknowledgements</td><td></td><td></td></tr>
<tr><td colspan="3">4. Abstract</td><td></td><td></td></tr>
<tr><td colspan="3">5. Table of Contents</td><td></td><td></td></tr>
<tr><td colspan="3">6. Main Content (Page numbers start from the main text)</td><td></td><td></td></tr>
<tr><td colspan="3">7. Reference</td><td></td><td></td></tr>
<tr><td colspan="3">8. Appendix (if need)</td><td></td><td></td></tr>
<tr><td colspan="3">9. Author’s Biography</td><td></td><td></td></tr>
<tr><td colspan="3">10. Task plan of senior thesis</td><td></td><td></td></tr>
<tr><td colspan="3">11. Assessment Form of senior thesis</td><td></td><td></td></tr>
<tr><td colspan="3">12. On-site Minutes Form of Oral Defense</td><td></td><td></td></tr>
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Abstract

Low-dose computed tomography (LDCT) reduces radiation exposure but introduces stronger quantum noise and streak-like artifacts, which may degrade anatomical boundaries and low-contrast structures. This thesis studies deep learning based LDCT denoising under a paired image restoration setting. Using the AAPM-Mayo 3mm B30 dataset, a complete experimental pipeline is built from paired slice preparation and patient-level splitting to training, tiled inference, quantitative evaluation, and residual analysis. RED-CNN is used as a convolutional baseline, CTformer is used as a Transformer-based CT denoising baseline, and a Restormer-style model named CTRestormer is adapted for LDCT restoration under local GPU memory constraints.

On the held-out L506 patient, the low-dose input obtains 29.2489 dB PSNR, 0.8759 SSIM, and 14.2416 RMSE. The best CTRestormer checkpoint at 21500 iterations improves the result to 32.6738 dB PSNR, 0.9096 SSIM, and 9.4842 RMSE, reaching performance comparable to RED-CNN and better than the reproduced CTformer baseline in this experiment. Beyond PSNR, SSIM, and RMSE, this thesis introduces residual noise analysis by comparing the estimated true residual, input minus target, with the residual removed by each model, input minus prediction. Residual correlation, removed residual strength, edge leakage, histograms, and frequency spectra are used to examine whether a model mainly removes noise-like texture or also suppresses structural information.

The main contribution of this thesis is therefore not only a denoising benchmark, but a practical and interpretable LDCT restoration workflow. The results show that CTRestormer is effective for LDCT denoising, while residual maps provide additional evidence for understanding denoising behavior and possible over-smoothing risk.

Keywords: Low-dose CT; image denoising; Transformer; CTRestormer; residual noise analysis; medical image restoration

Content

Chapter 1. Introduction

1.1 Research Background

Computed tomography (CT) is one of the most widely used medical imaging modalities because it provides high-resolution cross-sectional information about internal anatomical structures. It is used in diagnosis, screening, treatment planning, and follow-up examinations for many clinical tasks. However, CT imaging relies on X-ray radiation. Although the radiation dose of a single scan is usually controlled within clinical guidelines, cumulative exposure becomes a concern for repeated follow-up, pediatric imaging, population screening, and patients who require long-term monitoring. Reducing CT radiation dose while preserving image quality is therefore an important problem in medical imaging.

Low-dose CT (LDCT) imaging reduces the radiation burden by lowering the tube current, exposure time, or related acquisition settings. The cost of dose reduction is a higher noise level in the measured projection data and, after reconstruction, stronger noise and artifacts in the image domain. LDCT images often contain granular noise, streak-like patterns, and local texture degradation. These degradations can reduce the visibility of low-contrast lesions, make organ boundaries less reliable, and affect the confidence of image interpretation. Therefore, LDCT denoising is not merely a cosmetic image enhancement problem. It is a restoration task in which anatomical structure must be preserved while noise is suppressed.

Traditional denoising and reconstruction approaches include filtering, statistical iterative reconstruction, dictionary learning, and model-based regularization. These methods can be effective, but they often require careful parameter tuning and may struggle to balance noise removal with detail preservation. Deep learning has changed the landscape of medical image restoration. Convolutional neural networks (CNNs), such as RED-CNN, learn an image-domain mapping from LDCT to normal-dose CT (NDCT) and have become strong baselines for LDCT denoising [2]. More recently, Transformer-based models have been introduced into medical image denoising and general image restoration because attention mechanisms can model longer-range dependencies than local convolution alone [3, 4, 7].

However, higher PSNR or SSIM does not automatically mean that a denoised CT image is clinically safer. A model may improve global pixel metrics by producing a smoother image, but excessive smoothing can remove small structures or weaken edges. This is especially important for medical imaging, where the removed signal should be noise-like rather than anatomy-like. Motivated by this issue, this thesis evaluates denoising models not only through PSNR, SSIM, and RMSE, but also through residual noise maps. The central question becomes: what does the model remove from the LDCT input?

1.2 Research Objectives

This thesis studies supervised LDCT image denoising with paired LDCT and NDCT slices. The project has four main objectives. First, it aims to build a complete and reproducible LDCT denoising pipeline, including data preparation, patient-level splitting, model training, checkpoint management, full-slice inference, metric computation, and visualization. Second, it adapts a Restormer-style image restoration model to LDCT denoising and refers to this adapted model as CTRestormer. Third, it compares CTRestormer with a CNN baseline and a Transformer-based CT denoising baseline under the same evaluation setting. Fourth, it studies residual maps to interpret whether each model removes mainly dose-induced noise or also removes structure-related signal.

The task is formulated as paired image restoration. Given a low-dose CT slice x, a model f_θ predicts a denoised output $\hat{y} = f_\theta(x)$, which should approximate the normal-dose target y. The main performance metrics are PSNR, SSIM, and RMSE. In addition, this thesis defines the estimated LDCT residual as $x - y$, the removed residual as $x - \hat{y}$, and the prediction error as $\hat{y} - y$. These residual quantities provide a more detailed view of denoising behavior.

1.3 Main Contributions

The main contributions of this thesis are summarized as four parts.

First, this thesis constructs an end-to-end LDCT denoising workflow. The workflow starts from paired AAPM-Mayo LDCT and NDCT slices, uses a patient-level test split, trains multiple denoising models, performs full-resolution inference, computes quantitative metrics, and generates figures and residual maps. This makes the project more than an isolated model run; it is a complete experimental system.

Second, this thesis adapts a Restormer-style restoration Transformer to the LDCT denoising task. The proposed CTRestormer uses restoration-oriented Transformer blocks, hierarchical feature processing, residual prediction, patch-based training, and tiled inference. These design choices are important because full 512×512 CT slices are memory intensive for Transformer models. The implementation records practical strategies for running the model under limited local hardware.

Third, this thesis provides a fair comparison across models. RED-CNN is used as a representative CNN-based LDCT denoising baseline, while CTformer is used as a Transformer-based CT denoising baseline. CTRestormer is evaluated at several checkpoints to show training progression instead of relying on a single isolated result.

Fourth, this thesis introduces residual noise analysis as an additional interpretation layer. Instead of only looking at denoised images or aggregate metrics, the thesis compares what each model removes from the input. Residual correlation, edge leakage, histogram shape, and power spectrum are used to examine whether the removed residual resembles estimated LDCT noise or contains anatomical edges. This analysis provides evidence about possible over-smoothing risk, while still acknowledging that residual maps are not a substitute for clinical evaluation.

1.4 Thesis Organization

The remainder of the thesis is organized as follows. Chapter 2 reviews related work on LDCT denoising, CNN-based restoration, Transformer-based image restoration, and self-supervised denoising. Chapter 3 presents the methodology, including problem formulation, dataset preparation, baseline models, CTRestormer, and residual noise analysis. Chapter 4 describes the experimental setup, including the dataset split, training configuration, inference protocol, and evaluation metrics. Chapter 5 reports quantitative and qualitative results. Chapter 6 discusses residual noise analysis in detail and interprets model behavior beyond standard metrics. Chapter 7 concludes the thesis and outlines future work.

Chapter 2. Literature Review

2.1 Low-Dose CT Denoising

The motivation for LDCT denoising comes from the tension between radiation safety and image quality. Lower radiation dose reduces patient exposure, but it also increases the uncertainty of the measured signal. In reconstructed CT images, this uncertainty appears as noise, streak artifacts, and local texture distortion. LDCT denoising methods can be broadly grouped into projection-domain methods, reconstruction-domain methods, and image-domain post-processing methods. This thesis focuses on image-domain denoising, where the input is a reconstructed LDCT slice and the output is a restored image expected to resemble NDCT.

Image-domain denoising is attractive because it can be applied after reconstruction and can be evaluated directly with paired LDCT and NDCT slices. The AAPM-Mayo Low Dose CT Grand Challenge dataset has become a common resource for this purpose because it provides paired full-dose and quarter-dose CT data [1]. Under this paired setting, a supervised model can learn the mapping from noisy LDCT images to cleaner NDCT references. The limitation is that the NDCT target is still a clinical image rather than a mathematically noise-free ground truth. Therefore, the result should be interpreted as restoration toward the normal-dose reference, not as recovery of a perfect clean image.

Evaluation in LDCT denoising often uses PSNR, SSIM, and RMSE. PSNR and RMSE measure pixel-level fidelity, while SSIM measures structural similarity [9]. These metrics are useful for reproducible comparison, but they do not fully describe clinical usefulness. For example, a strongly smoothed image may have a lower pixel error but weaker fine structures. This limitation motivates the residual noise analysis used later in this thesis.

2.2 CNN-Based Image Restoration

CNNs have been highly influential in medical image restoration. Their local receptive fields and shared weights make them effective at learning spatial patterns such as edges, textures, and local noise statistics. RED-CNN is a representative LDCT denoising method that uses a residual encoder-decoder structure to restore CT images [2]. It remains a meaningful baseline because it was designed specifically for LDCT and has demonstrated strong performance on paired CT denoising tasks.

CNN-based models have several advantages. They are relatively stable to train, computationally efficient compared with many attention-based models, and well matched to local image restoration. In CT denoising, local convolution can remove high-frequency noise while preserving many anatomical boundaries. U-Net style encoder-decoder structures further improve multi-scale feature extraction and have influenced many biomedical imaging models [6].

However, CNNs also have limitations. A standard convolution only aggregates information within a local neighborhood. Although deeper networks, dilated convolutions, skip connections, and multi-scale designs can enlarge the effective receptive field, long-range dependency modeling is still indirect. In LDCT images, noise and artifacts may interact with anatomical structures across larger regions, especially when streak artifacts are present. CNN models may also produce over-smoothed outputs if the loss function and architecture encourage average-looking predictions. For these reasons, Transformer-based restoration models are increasingly studied as alternatives or complements.

2.3 Transformer-Based Restoration

The Transformer was originally introduced for sequence modeling and uses attention to model relationships between tokens [7]. Vision Transformer later showed that attention-based architectures could be applied to images by representing an image as a sequence of patches [8]. For image restoration, the challenge is that restoration requires dense pixel-level output and often high-resolution input. Direct global self-attention can be expensive for large images, so efficient attention mechanisms and hierarchical designs are important.

SwinIR applies shifted-window attention to image restoration and demonstrates that Transformer architectures can be competitive in tasks such as denoising and super-resolution [5]. Restormer further designs an efficient Transformer for high-resolution image restoration, using restoration-oriented blocks and a hierarchical structure to reduce memory cost [4]. These ideas are relevant to LDCT denoising because CT slices are high-resolution grayscale images and require preservation of fine anatomical structures.

CTformer is a Transformer-based model designed for LDCT denoising [3]. It shows that attention-based methods can be adapted to CT-specific restoration. This thesis uses CTformer as a Transformer baseline and adapts a Restormer-style model as CTRestormer. The comparison is useful because CTformer represents a CT-specific Transformer design, while CTRestormer represents a general image restoration Transformer adapted to the LDCT setting.

2.4 Literature Gap and Thesis Position

The reviewed literature suggests three gaps that motivate this thesis. First, CNN-based methods are strong but may not explicitly model long-range dependencies. Second, Transformer-based restoration can improve contextual modeling, but high-resolution CT slices create practical memory challenges. Third, many denoising studies focus on PSNR, SSIM, and RMSE, while providing less analysis of what signal is removed from the noisy input.

This thesis is positioned at the intersection of these gaps. It keeps RED-CNN as a strong CNN baseline, evaluates CTformer as a CT-specific Transformer baseline, and adapts a Restormer-style architecture to LDCT denoising. More importantly, it adds residual noise analysis to examine denoising behavior. By comparing $x-y$ with $x-\hat{y}$, the thesis studies whether the removed residual resembles estimated LDCT noise. This does not prove clinical safety, but it provides a useful interpretation beyond global metrics.

Chapter 3. Methodology

3.1 Problem Formulation

This thesis formulates LDCT denoising as a paired supervised image restoration problem. Let $x \in \mathbb{R}^{H\times W}$ denote a low-dose CT slice and $y \in \mathbb{R}^{H\times W}$ denote the corresponding normal-dose CT slice. A denoising model f_θ predicts

$$\hat{y} = f_\theta(x), \tag{3-1}$$

where θ represents learnable parameters. The goal is to make $\hat{y}$ close to y, while preserving anatomical structures in x.

The training objective follows the general supervised restoration setting. The model is optimized on paired LDCT and NDCT patches. Patch-based training is used because full CT slices are memory intensive, especially for Transformer-based restoration models. During evaluation, the model is applied to complete 512×512 slices through tiled inference so that final metrics are computed on full images rather than only on training patches.

3.2 Dataset and Preprocessing

The experiments use paired slices prepared from the AAPM-Mayo LDCT dataset [1]. The working dataset contains 3mm B30 LDCT and NDCT image pairs from ten patients: L067, L096, L109, L143, L192, L286, L291, L310, L333, and L506. Patient L506 is held out as the test patient. This patient-level split is important because slice-level random splitting can place neighboring slices from the same patient in both training and testing, which may overestimate generalization.

The preprocessing pipeline starts from paired DICOM series. For each patient and dose level, slices are sorted by their axial position, converted from stored pixel values to Hounsfield units (HU) using the DICOM rescale slope and intercept, and saved as paired NumPy arrays. Invalid background values marked as -2000 in the original conversion are set to zero before HU conversion. The HU images are then linearly normalized from $[-1024, 3072]$ to $[0, 1]$. No CT display windowing is applied before training; the display window $[-160, 240]$ HU is used only after denormalization for visualization and metric preparation.

Each full slice has spatial size 512×512. During training, the data loader samples paired 64×64 crops from the same spatial location in the LDCT and NDCT images. The same flip or rotation augmentation is applied to both images in a pair, so the supervised target remains aligned with the low-dose input. The 64×64 size is therefore a memory-aware training crop size, not the original image size and not a ViT-style token size. During testing, the model is evaluated on full slices by tiled inference, and the reported metrics are computed on reconstructed 512×512 predictions.

3.3 Baseline Models

Three types of inputs or models are compared in this thesis. The first is the raw low-dose input, which provides a lower-bound reference. The second is RED-CNN, a CNN-based LDCT denoising model [2]. RED-CNN is used because it is a widely recognized baseline for CT denoising and represents a strong convolutional approach. The third is CTformer, a Transformer-based model designed for LDCT denoising [3]. CTformer is included to compare the proposed CTRestormer with another attention-based CT denoising method.

Using these baselines makes the evaluation more informative. If CTRestormer is compared only with the low-dose input, improvement is expected and not very meaningful. Comparing it with RED-CNN and CTformer shows whether the adapted model is competitive with both CNN and Transformer baselines. This also helps interpret residual behavior: a method with similar PSNR may remove different types of residual signal.

3.4 Proposed CTRestormer

CTRestormer is the Restormer-style model adapted in this thesis for LDCT denoising. It is not presented as a new theory of attention, but as a practical adaptation of

Table 3.1 Dataset split and evaluation protocol.

Item	Setting	Purpose
Dataset	AAPM-Mayo 3mm B30 paired LDCT/NDCT slices	Supervised low-dose to normal-dose CT restoration
Patients	L067, L096, L109, L143, L192, L286, L291, L310, L333, L506	Patient-level organization of paired slices
Test patient	L506	Held-out evaluation to reduce patient-level leakage
Slice count	2167 training slices and 211 held-out L506 test slices	Explicit patient-level split size
Image size	512×512	Full CT slice denoising and tiled inference
Preprocessing	DICOM to HU, paired NumPy saving, normalization from $[-1024, 3072]$ to $[0, 1]$	Stable numerical range for supervised training
Training samples	Random paired 64×64 crops with matched augmentation	Memory-aware training while preserving LDCT/NDCT alignment
Metrics	PSNR, SSIM, RMSE, residual correlation, edge leakage	Image quality and residual behavior evaluation

high-resolution Transformer restoration to CT images. The model follows the idea that image restoration should preserve spatial detail while learning the degradation-specific correction. Compared with classification-oriented Transformers, restoration Transformers must output dense images and maintain local alignment.

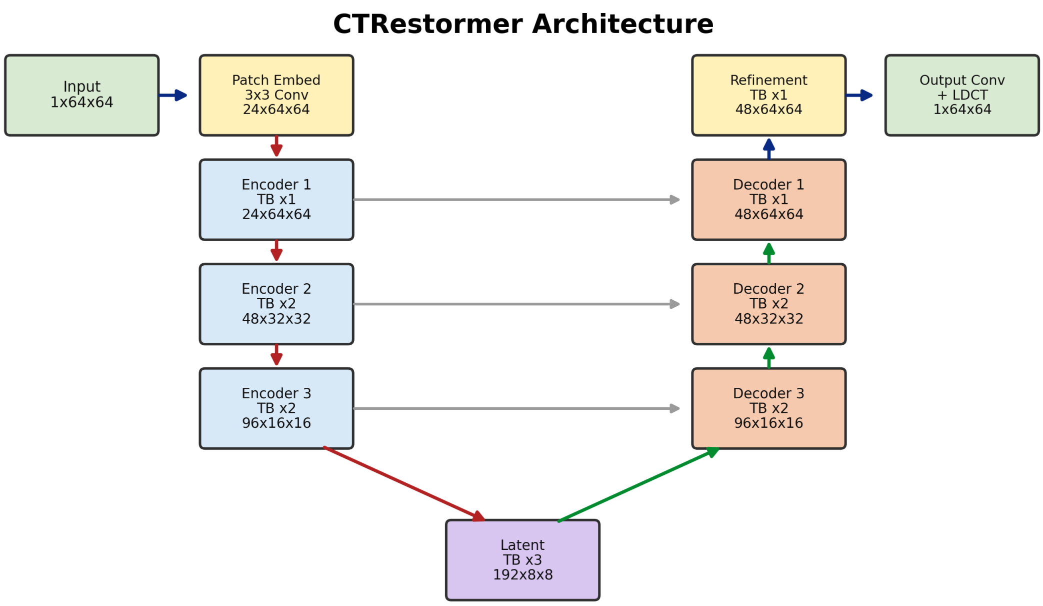

Figure 3.1 Compact CTRestormer architecture used in this thesis. Red arrows indicate downsampling, green arrows indicate upsampling, and gray arrows indicate encoder-decoder skip connections. The highlighted dimensions show the feature size at each stage.

The proposed pipeline uses overlap patch embedding, hierarchical feature processing, Transformer restoration blocks, skip-connected decoding, refinement blocks, and residual output prediction, as shown in Figure 3.1. The actual trained configuration uses base dimension 24. For a 64×64 training crop, the model input tensor is $1 \times 64 \times 64$. The first model layer is an overlap patch embedding layer implemented as a 3×3 convolution with stride 1 and padding 1. It maps the input to a $24 \times 64 \times 64$ feature map while keeping the spatial resolution unchanged.

This step should not be confused with ViT-style patch tokenization. In ViT, an image is commonly divided into non-overlapping patches and converted into a token sequence before Transformer processing. In the CTRestormer implementation used here, the 64×64 crop remains a two-dimensional feature map. The patch embedding is a convolutional feature projection, not a preprocessing step outside the network. The feature map then passes through three downsampling stages to $48 \times 32 \times 32$, $96 \times 16 \times 16$, and $192 \times 8 \times 8$. The decoder reconstructs high-resolution features with PixelShuffle upsampling, skip fusion, and channel reduction. After the final skip concatenation, the high-resolution decoder and refinement stages operate on $48 \times 64 \times 64$ features. The final output convolution predicts a residual correction that is added to the LDCT input.

Each Transformer restoration block (TB) contains two residual sub-blocks, as shown in Figure 3.2. The first sub-block applies two-dimensional layer normalization followed by an MDTA-style attention module. In this implementation, query, key, and value features are generated by 1×1 convolution and depthwise 3×3 convolution, so the attention branch still contains local spatial mixing. The second sub-block applies another normalization and a gated depthwise feed-forward network (GDFN). A TB keeps the spatial size unchanged; only the channel count C depends on the current encoder or decoder level. This combination is useful for image restoration because it combines attention-based feature interaction with convolution-like local bias.

In Figure 3.2, LN2d denotes two-dimensional layer normalization applied on feature maps. Unlike the layer normalization used for one-dimensional token sequences, LN2d normalizes the channel features while preserving the $H \times W$ spatial layout. MDTA denotes the attention branch, where query, key, and value features are produced by convolutional projections. GDFN denotes the gated depthwise feed-forward network, which expands channels, applies depthwise spatial mixing, and uses a gated nonlinearity before projecting features back. The $+$ symbols denote residual addition: the block adds the branch output back to its input feature map. These residual connections help stabilize training and preserve anatomical information by allowing the model to learn feature corrections rather than replacing the whole representation.

The architecture is effective for LDCT denoising for several reasons. First, overlap patch embedding avoids hard patch boundaries and preserves local continuity in CT

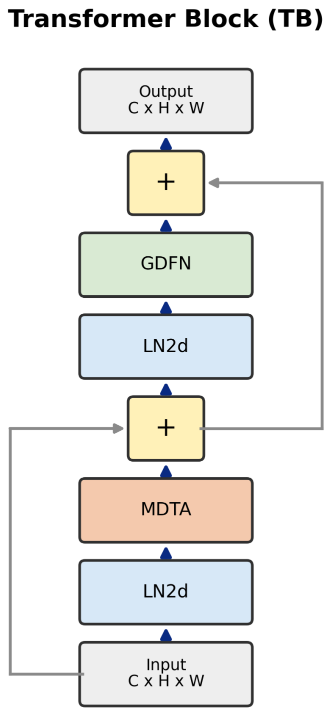

Figure 3.2 Transformer restoration block used in CTRestormer. Each TB keeps the input feature size $C \times H \times W$, applies an attention residual branch and a gated feed-forward residual branch, and preserves spatial alignment for dense image restoration.

images. Second, the encoder-decoder hierarchy lets the model combine fine local texture with larger contextual information. Third, skip connections pass high-resolution structure from the encoder to the decoder, which helps preserve anatomical boundaries. Fourth, residual output prediction constrains the model to learn a correction between LDCT and NDCT instead of generating a completely new image. Finally, tiled inference makes the model practical for full 512×512 CT slices even though full-image attention would exceed memory limits.

Compared with RED-CNN, CTRestormer has stronger long-range feature interaction. RED-CNN is stable and efficient, but its convolutional layers mainly aggregate local neighborhoods and do not explicitly model global dependencies. This can limit its ability to distinguish structured artifacts from anatomical texture. Compared with the local CTformer implementation, CTRestormer has a more restoration-oriented hierarchy. CTformer uses token-to-token encoding and a shallow Transformer bottleneck in this project, which is lightweight but less expressive for multi-scale image restoration. CTRestormer adds a U-shaped reconstruction path, explicit skip fusion, deeper restoration blocks, and residual prediction; these choices make it more suitable for dense denoising.

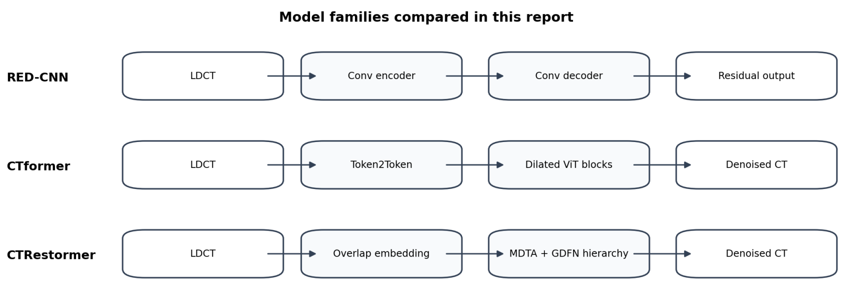

Figure 3.3 Schematic comparison of the evaluated denoising models. RED-CNN represents a convolutional baseline, CTformer represents a CT-specific Transformer baseline, and CTRestormer is the Restormer-style model adapted in this thesis.

3.5 Memory-Aware Training and Tiled Inference

High-resolution Transformer restoration can be limited by GPU memory. A full 512×512 CT slice contains many spatial tokens, and attention-based modules can be expensive. Therefore, the implementation uses memory-aware training and inference strategies. During training, patches are sampled from paired slices. Patch-based training reduces memory usage and increases the number of local training samples. Mixed precision and gradient accumulation can further improve practical feasibility on limited hardware.

During inference, tiled prediction is used to reconstruct full slices. In the current implementation, a 512×512 slice is padded and divided into overlapping 64×64 tiles with stride 32. Each tile is processed by the model, and the central regions of the tile predictions are aggregated into a full 512×512 output. Tiled inference avoids out-of-memory errors while still allowing metrics to be computed on complete slices. This engineering detail is important because the goal of LDCT denoising is full-image restoration, not only patch-level prediction.

3.6 Residual Noise Analysis

To analyze what each model removes, this thesis defines three residual quantities:

$$r_{\text{true}} = x - y, \tag{3-2}$$

$$r_{\text{removed}} = x - \hat{y}, \tag{3-3}$$

$$e_{\text{pred}} = \hat{y} - y. \tag{3-4}$$

Here r_{true} is the estimated LDCT residual relative to the NDCT target, r_{removed} is the residual removed by the model, and e_{pred} is the remaining prediction error. If a model removes noise in a desirable way, r_{removed} should share texture and distributional similarity with r_{true}, while containing limited anatomical edge information.

The first residual metric is residual correlation:

$$\rho = \operatorname{corr}(r_{\text{true}}, r_{\text{removed}}). \tag{3-5}$$

A higher correlation suggests that the removed residual is more aligned with the estimated LDCT noise, although this metric alone is not sufficient to prove clinical safety. The second metric is the standard deviation of the removed residual, which measures removal strength. The third metric is edge leakage. It estimates how strongly the removed residual overlaps with edge-like structures. Higher edge leakage indicates a stronger risk that the model removes anatomical boundary information together with noise.

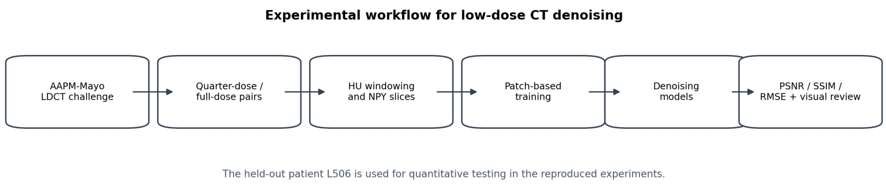

Figure 3.4 Experimental workflow from paired LDCT/NDCT preparation to model training, full-slice inference, metric evaluation, and residual analysis.

Chapter 4. Experimental Setup

4.1 Implementation Environment

The project is implemented as a local LDCT denoising workflow. The codebase prepares paired NumPy arrays, trains denoising models, saves checkpoints, performs inference on the held-out patient, and generates figures for thesis analysis. The experiments are designed around practical constraints. Transformer-based restoration models can exceed memory limits when applied directly to full CT slices, so patch training and tiled inference are used.

All reported local experiments were run on the workstation summarized in Table 4.1. The hardware is a practical single-GPU student research environment rather than a large training server. This constraint affects the experimental design: full 512×512 slice training is avoided for Transformer-based models, and patch-based training with tiled full-slice inference is used to keep GPU memory usage feasible.

Table 4.1 Local implementation environment used for the reported experiments.

Item	Configuration
CPU	AMD Ryzen 7 5800H, 8 cores and 16 logical processors
Memory	Approximately 14.9 GB physical memory
GPU	NVIDIA GeForce RTX 3060 Laptop GPU
CUDA	CUDA 12.8
Python and PyTorch	Python 3.8.20; PyTorch 2.4.1 with CUDA 12.1 build

4.2 Training and Inference Configuration

Table 4.2 summarizes the model and inference configuration used in the current thesis. The purpose of this table is to make the experimental protocol easier to audit. It separates model identity, training strategy, and inference strategy from the results reported later.

Table 4.2 Model and inference configuration summary.

Component	Configuration	Reason
RED-CNN	CNN denoising baseline	Representative convolutional LDCT denoiser
CTformer	Transformer LDCT baseline checkpoint	Comparison with a CT-specific Transformer model
CTRestormer	Restormer-style encoder-decoder with residual output	Main adapted restoration model for LDCT
Training	Patch-based training with memory-aware settings	Enables training under local GPU limitations
Inference	Tiled full-slice inference	Avoids out-of-memory errors for 512×512 slices

4.3 Model Size

Table 4.3 reports the parameter sizes of the main models used in this thesis. The values were computed by instantiating the exact model definitions used in the experiments and counting trainable parameters. CTformer contains a fixed sinusoidal positional embedding, so its total parameter count is larger than its trainable parameter count. CTRestormer uses the actual lightweight configuration trained in this thesis, not the larger default class configuration.

Table 4.3 Parameter sizes of the evaluated denoising models.

Model	Main configuration	Total params	Trainable params
RED-CNN	out_ch=96	1,848,865	1,848,865
CTformer	embed=64, depth=1, heads=8	1,652,493	1,447,477
CTRestormer	dim=24, blocks=(1,2,2,3)	2,985,981	2,985,981

The proposed CTRestormer is larger than RED-CNN and CTformer in the current implementation, but it remains small enough for local patch-based training. The extra capacity is used in the multi-scale encoder-decoder and Transformer restoration blocks. This is a deliberate trade-off: CTRestormer is not the lightest model, but it provides stronger representation capacity and better restoration bias for LDCT denoising.

4.4 Evaluation Metrics

Three standard image quality metrics are used for the main comparison. PSNR measures the peak signal-to-noise ratio and is reported in decibels. Higher PSNR indicates lower pixel-level error relative to the target. SSIM measures structural similarity and is also higher-is-better [9]. RMSE is the root mean square error between the prediction and target; lower RMSE indicates better fidelity.

These metrics are computed on the held-out L506 patient. The low-dose input is included in the metric table to show the improvement produced by learned denoising. For CTRestormer, several checkpoints are included to show how performance changes during training. In addition to the standard metrics, residual correlation, removed residual standard deviation, removed residual MAE, prediction error MAE, and edge leakage are computed for the representative residual analysis slice.

4.5 Experimental Questions

The experiments are designed to answer four questions. First, do learned denoising models improve LDCT image quality compared with the raw low-dose input? Second, how does CTRestormer compare with RED-CNN and CTformer on the held-out patient? Third, does the CTRestormer checkpoint study show consistent training improvement? Fourth, what residual signal does each model remove from the low-dose input?

The fourth question is important because the best metric value is not the only concern in medical image denoising. If a model removes residual patterns that contain clear anatomical edges, the denoised image may look clean but lose information. Therefore, residual analysis is treated as a separate discussion chapter rather than a minor visual appendix.

Chapter 5. Experimental Results

5.1 Quantitative Comparison

Table 5.1 reports the main quantitative results on the held-out L506 patient. The low-dose input has 29.2489 dB PSNR, 0.8759 SSIM, and 14.2416 RMSE. All learned models substantially improve the low-dose input, confirming that supervised denoising is effective under the current paired setting. The CTformer result is reported with the updated 100000-iteration continuation checkpoint.

Table 5.1 Quantitative results on the held-out L506 test patient.

Method	Checkpoint	PSNR ↑	SSIM ↑	RMSE ↓
Low-dose input	–	29.2489	0.8759	14.2416
CTformer	54718	32.3852	0.9026	9.8366
CTformer	100000	32.6688	0.9067	9.5197
RED-CNN	9500	32.6656	0.9067	9.4867
CTRestormer	11500	32.1294	0.8997	10.0689
CTRestormer	16500	32.3584	0.9049	9.8145
CTRestormer	21500	**32.6738**	**0.9096**	**9.4842**

The updated CTformer 100000 checkpoint reaches 32.6688 dB PSNR, 0.9067 SSIM, and 9.5197 RMSE. Compared with RED-CNN 9500, it is slightly higher in PSNR by 0.0032 dB and has the same four-decimal SSIM, while RED-CNN remains slightly better in RMSE by 0.0330. The best CTRestormer checkpoint still gives the best overall metrics in this table, reaching 32.6738 dB PSNR, 0.9096 SSIM, and 9.4842 RMSE. Compared with CTformer at the same 21500-iteration checkpoint, CTRestormer improves PSNR by 0.8279 dB, improves SSIM by 0.0085, and reduces RMSE by 1.0418.

5.2 Estimated Training-Budget Comparison

Iteration count is not a fair proxy for computational budget because the three models have different architectures, different per-iteration costs, and different training code. The raw elapsed fields in the historical logs also cannot be directly accumulated because several runs were interrupted and later resumed. To still provide a useful training-

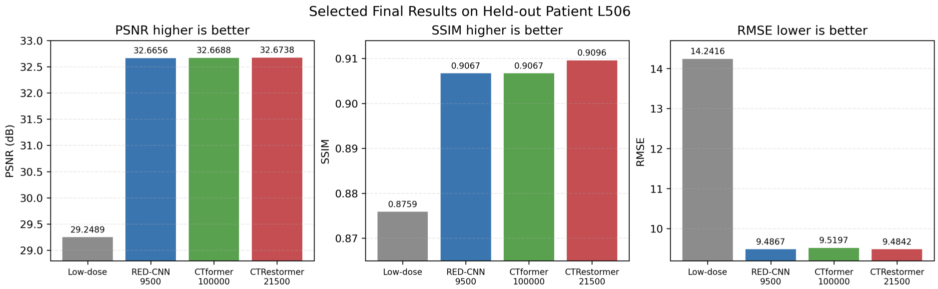

Figure 5.1 Metric comparison of selected final results on the held-out L506 patient. CTformer uses the updated 100000-iteration checkpoint.

budget reference, this thesis estimates a 100-minute training budget from continuous checkpoint-saving segments that show no obvious long interruption. This estimate is not a strict same-wall-clock experiment, but it is more informative than comparing only final selected checkpoints.

Table 5.2 shows the speed-estimation protocol. For each model, a representative continuous checkpoint segment is used to estimate iterations per minute. The estimated 100-minute iteration count is then rounded to the nearest available or evaluated checkpoint. For CTformer, the 70000 checkpoint should be read as a 100-minute-equivalent checkpoint estimated from the observed checkpoint-saving speed, not as proof that the whole training history up to 70000 iterations took only 100 minutes.

Table 5.2 Estimated 100-minute training budget from representative continuous checkpoint segments.

Model	Segment time	Iter/min	Estimated 100-min iter	Used checkpoint
CTformer	49.6 min	705.1	70505	70000
RED-CNN	95.9 min	93.8	9380	9500
CTRestormer	80.0 min	137.5	13754	14000

Table 5.3 reports the corresponding test metrics. Under this estimated 100-minute budget, RED-CNN gives the best PSNR and RMSE. CTRestormer is slightly higher than CTformer in SSIM, but its PSNR and RMSE are not better at this estimated early budget. This result is useful because it prevents overstating the proposed model: CTRestormer is not the most efficient short-budget model in the current implementation. Its advantage appears more clearly after longer training, as shown by the final 21500-iteration

checkpoint in Table 5.1.

Table 5.3 Estimated 100-minute training-budget comparison on L506.

Model	Used checkpoint	Approx. epoch	PSNR ↑	SSIM ↑	RMSE ↓
CTformer	70000	129.2	32.4017	0.9012	9.8020
RED-CNN	9500	69.9	**32.6656**	**0.9067**	**9.4867**
CTRestormer	14000	12.9	32.3718	0.9044	9.8020

Table 5.4 reports a same-iteration comparison between CTformer and CTRestormer at 21500 iterations because both checkpoints exist in the current experiment records. This table is useful for showing relative progress at the same checkpoint index, but it should not be interpreted as a same-compute comparison. RED-CNN is not included in this table because its saved checkpoint schedule and training code are different.

Table 5.4 Same-iteration comparison between CTformer and CTRestormer on L506.

Model	Iteration	PSNR ↑	SSIM ↑	RMSE ↓
CTformer	21500	31.8459	0.9011	10.5260
CTRestormer	21500	**32.6738**	**0.9096**	**9.4842**

This comparison shows that CTRestormer is much stronger than CTformer at the same iteration count. The reason is not only parameter count: CTRestormer has a restoration-oriented multi-scale encoder-decoder, skip fusion, local convolutional mixing inside attention and feed-forward branches, and residual output prediction. CTformer becomes competitive only after much longer continuation training to 100000 iterations, while CTRestormer reaches comparable or better metrics at 21500 iterations. RED-CNN remains a strong baseline, but its training loop and saved checkpoint schedule are different, so it is more appropriately discussed in the selected-checkpoint comparison in Table 5.1 and the estimated time-budget comparison in Table 5.3.

5.3 Qualitative Comparison

Figure 5.2 shows a representative qualitative result for slice L506_88. Qualitative comparison is necessary because PSNR and SSIM do not directly reveal texture quality, boundary sharpness, or possible over-smoothing. The CTformer panel uses the updated 100000-iteration checkpoint. The low-dose input contains stronger noise texture, while the denoised outputs are visually closer to the normal-dose target. The comparison should be interpreted together with residual maps in Chapter 6.

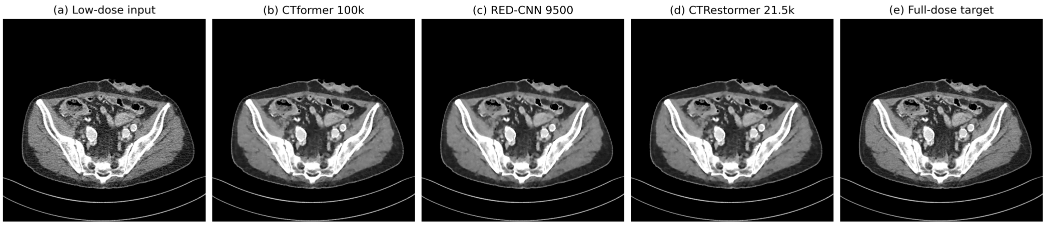

Figure 5.2 Representative qualitative comparison for slice L506_88. All panels are shown under the same display setting.

Two additional slices are included to make the visual comparison less dependent on a single representative image. Slice L506_45 was selected from an earlier axial range with relatively high input-target error within that range, while slice L506_180 has the highest input-target RMSE among the scanned L506 slices. These two cases cover different anatomical regions and make denoising differences easier to inspect.

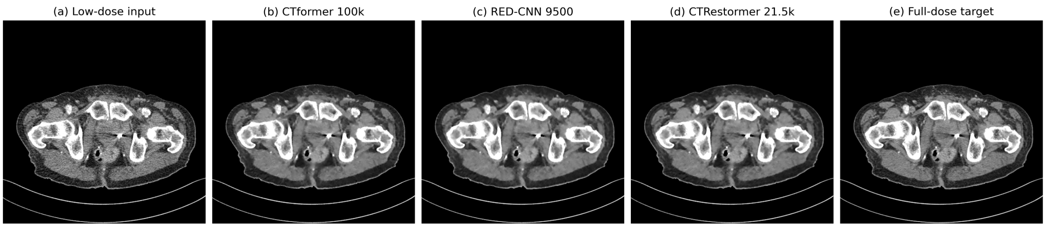

Figure 5.3 Additional qualitative comparison for slice L506_45. This slice shows a pelvic region and provides a separate visual case from the abdominal slice used in Figure 5.2.

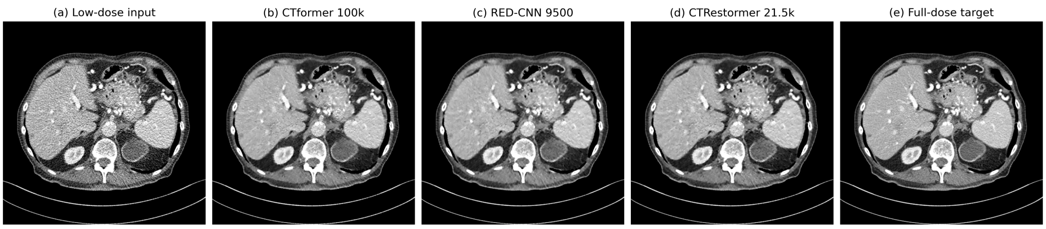

Figure 5.4 Additional qualitative comparison for slice L506_180. This slice has the highest low-dose to full-dose RMSE among the inspected L506 slices, so it is useful for comparing denoising behavior under stronger visible noise.

5.4 Training Curve

Figures 5.5 show the logged training loss trends for CTRestormer.

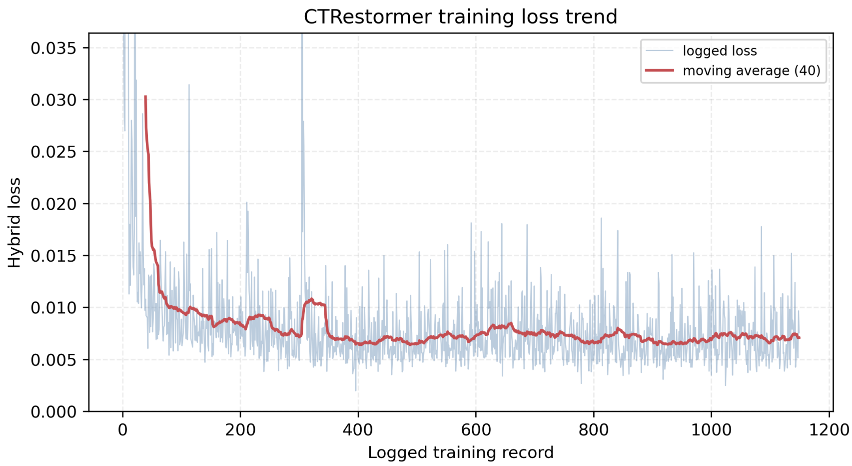

Figure 5.5 Training loss trend from the CTRestormer run. The plotted loss is the logged loss.

Chapter 6. Residual Noise Analysis and Discussion

6.1 Motivation

The main quantitative results show that CTRestormer 21500 achieves strong PSNR, SSIM, and RMSE. However, medical image denoising should not be judged only by aggregate image quality metrics. A denoised image can look cleaner because noise has been removed, but it can also look cleaner because the model has suppressed subtle anatomical texture. For this reason, this thesis studies residual maps.

The residual analysis asks a direct question: what did the model remove from the low-dose input? The estimated true residual is $x-y$, where x is the LDCT input and y is the NDCT target. The removed residual is $x-\hat{y}$, where $\hat{y}$ is the model prediction. If the removed residual resembles the estimated true residual and contains limited edge-like anatomical structure, the model is more likely to be removing noise rather than simply smoothing the image.

6.2 Residual Map Results

Figure **??** presents the standardized prediction comparison for L506_88. The CTformer prediction in this residual study uses the updated 100000-iteration checkpoint. The figure separates the visual comparison from the residual analysis so that the reader can first inspect image-level restoration quality. Figure **??** then focuses only on the residual removed by each model. This separation is important because a denoised image may look acceptable even when the removed residual contains structure-related signal.

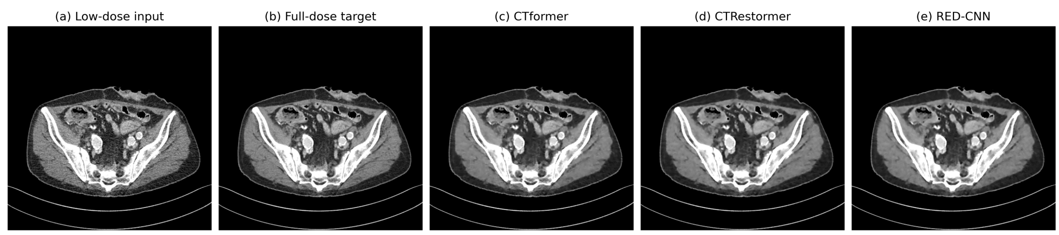

Figure 6.1 Standardized prediction comparison for L506_88 using a shared CT display window. CTformer uses the updated 100000-iteration checkpoint.

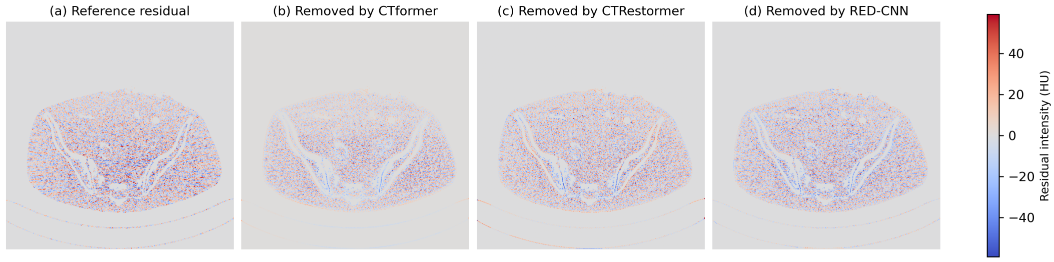

Figure 6.2 Removed residual maps computed as input minus prediction. The panels use a shared diverging residual scale to compare removal strength.

The residual maps show different denoising behaviors. CTformer removes a more conservative residual, while CTRestormer and RED-CNN remove stronger residual patterns. Stronger removal is not automatically better; it must be interpreted together with prediction error and edge leakage. If the removed residual contains obvious anatomical edges, the denoising model may be suppressing structure. If the removed residual is mainly noise-like texture, the behavior is more desirable.

6.3 Prediction Error and Residual Statistics

Figure **??** shows the prediction error maps for L506_88. These maps represent $\hat{y}-y$, so they show what remains different from the NDCT target after denoising. Figure **??** summarizes residual correlation and edge leakage. Table **??** reports the residual statistics.

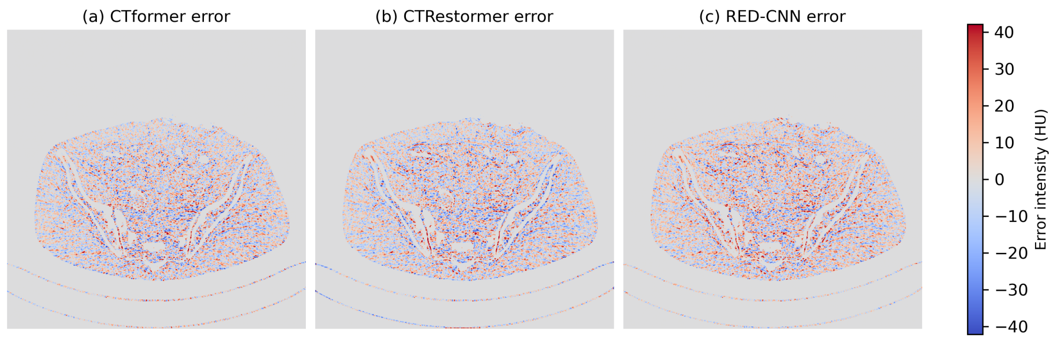

Figure 6.3 Prediction error maps computed as prediction minus target for L506_88.

The three models have residual correlations around 0.72 to 0.74, suggesting that their removed residuals are substantially related to the estimated LDCT residual. CT-

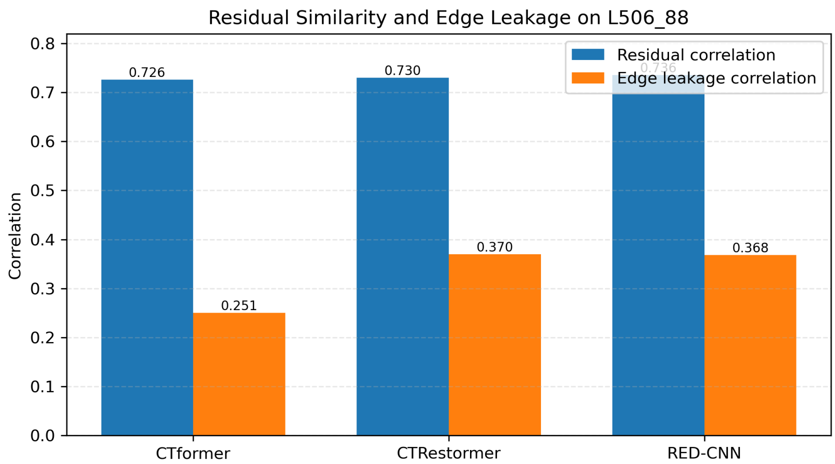

Figure 6.4 Residual correlation and edge leakage comparison for the representative L506_88 slice.

Table 6.1 Residual noise statistics for representative slice L506_88.

Model	Corr.	Edge	Std	Removed MAE	Error MAE
CTformer 100000	0.7259	0.2506	9.8406	4.3490	4.1493
CTRestormer	0.7300	0.3699	11.3863	5.2081	4.1636
RED-CNN	0.7358	0.3683	11.7800	5.3967	4.1823

former has the lowest edge leakage, which suggests more conservative residual removal. CTRestormer and RED-CNN remove stronger residuals and have similar edge leakage values. This supports a balanced interpretation: CTRestormer achieves strong image quality metrics, but its residual behavior should still be inspected for possible structure suppression.

6.4 Histogram and Frequency Analysis

Residual histograms and power spectra provide complementary information. The histogram shows the distribution of removed residual values, while the power spectrum shows how residual energy is distributed across spatial frequencies. Figures **??** and **??** summarize these two views for L506_88.

These plots help distinguish denoising strength from denoising quality. A method that removes more residual energy may produce a smoother output, but whether the removed energy is desirable depends on its spatial and structural characteristics. Frequency analysis is therefore useful when comparing conservative and aggressive denois-

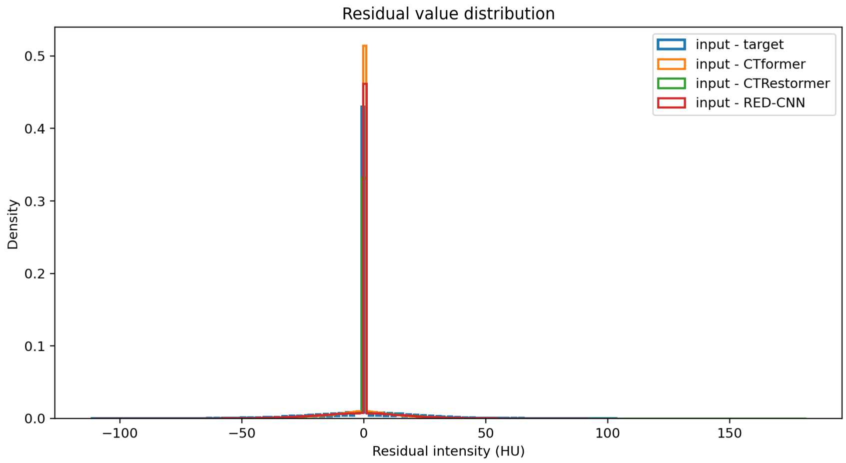

Figure 6.5 Histogram of removed residual values for L506_88.

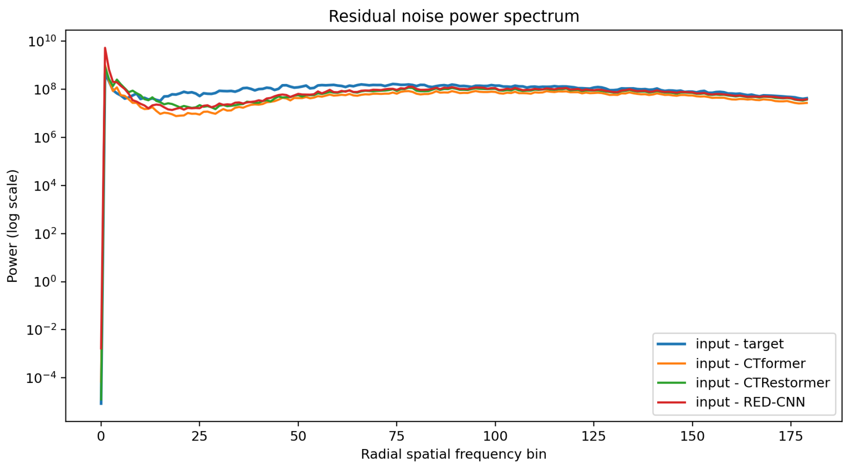

Figure 6.6 Radially averaged residual power spectrum for L506_88.

ing behavior.

6.5 Limitations of the Residual Study

The current residual analysis is a representative case study on slice L506_88. It provides useful evidence, but it should not be overstated as proof over the entire dataset. A stronger final study should compute residual correlation, removed residual standard deviation, and edge leakage for all slices in the held-out L506 patient and report mean and standard deviation.

Another limitation is that $x - y$ is only an approximation of true LDCT noise. The NDCT target is used as a reference, but it is not a perfect noise-free image. In addition, residual maps do not directly measure clinical diagnostic accuracy. A future clinical-quality study would require region-of-interest measurements, reader studies, or task-based evaluation. Therefore, this thesis uses residual analysis as model-behavior evidence, not as a direct clinical safety claim.

Chapter 7. Conclusion and Future Work

7.1 Conclusion

This thesis studied Transformer-based LDCT image denoising using a paired supervised restoration setting. A complete experimental pipeline was built using AAPM-Mayo 3mm B30 paired slices, with patient L506 held out for testing. RED-CNN was used as a CNN baseline, CTformer was used as a Transformer-based CT denoising baseline, and CTRestormer was adapted from Restormer-style image restoration for LDCT denoising.

The quantitative results show that all learned models improve the raw low-dose input. After continued training, the updated CTformer 100000 checkpoint reaches 32.6688 dB PSNR, 0.9067 SSIM, and 9.5197 RMSE, slightly exceeding RED-CNN in PSNR and matching it in SSIM while remaining slightly worse in RMSE. The best CTRestormer checkpoint, at 21500 iterations, achieves 32.6738 dB PSNR, 0.9096 SSIM, and 9.4842 RMSE on the held-out L506 patient. This is the best overall result among the selected checkpoints in this experiment. The checkpoint comparisons also show that both CTformer continuation and CTRestormer training progression improve performance relative to earlier checkpoints.

Beyond standard metrics, this thesis introduced residual noise analysis. By comparing input minus target with input minus prediction, the thesis examined what signal each model removes. The residual analysis shows that CTformer is more conservative, while CTRestormer and RED-CNN remove stronger residuals. CTRestormer obtains strong image quality metrics, but its edge leakage is higher than CTformer on the representative slice, so structure preservation should remain part of future evaluation.

Overall, the contribution of this thesis is it combines model comparison, memory-aware Transformer adaptation, tiled inference, quantitative evaluation, qualitative visualization, and residual behavior analysis.

7.2 Future Work

Several extensions can improve the thesis and the research project. First, residual metrics should be computed over all L506 test slices. Reporting mean and standard deviation would make the residual analysis more reliable. Second, region-of-interest evaluation should be added for soft tissue, bone boundaries, and low-contrast regions. Such analysis would better connect image restoration metrics to clinically relevant structures. Third, more studies can be performed for CTRestormer. Patch size, loss function, attention block configuration, and inference tile overlap may all affect denoising behavior. Fourth, a pseudo Noise2Noise or self-supervised denoising experiment can be explored. This would be useful in settings where paired LDCT and NDCT data are unavailable.

References

[1] American Association of Physicists in Medicine. Low Dose CT Grand Challenge, 2016. Available: https://www.aapm.org/GrandChallenge/LowDoseCT/.

[2] H. Chen, Y. Zhang, M. K. Kalra, F. Lin, Y. Chen, P. Liao, J. Zhou, and G. Wang, "Low-dose CT with a residual encoder-decoder convolutional neural network," *IEEE Transactions on Medical Imaging*, vol. 36, no. 12, pp. 2524–2535, 2017.

[3] D. Wang, F. Fan, Z. Wu, R. Liu, F. Wang, and H. Yu, "CTformer: Convolution-free token2token dilated vision Transformer for low-dose CT denoising," *Physics in Medicine and Biology*, vol. 68, no. 6, 2023.

[4] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, M.-H. Yang, and L. Shao, "Restormer: Efficient Transformer for high-resolution image restoration," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 5728–5739.

[5] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, "SwinIR: Image restoration using Swin Transformer," in *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, 2021, pp. 1833–1844.

[6] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention*, 2015, pp. 234–241.

[7] A. Vaswani et al., "Attention is all you need," in *Advances in Neural Information Processing Systems*, 2017, pp. 5998–6008.

[8] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *International Conference on Learning Representations*, 2021.

[9] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.

[10] J. Lehtinen et al., "Noise2Noise: Learning image restoration without clean data," in *Proceedings of the International Conference on Machine Learning*, 2018.

[11] J. Batson and L. Royer, "Noise2Self: Blind denoising by self-supervision," in *Proceedings of the International Conference on Machine Learning*, 2019.

[12] A. Krull, T.-O. Buchholz, and F. Jug, "Noise2Void: Learning denoising from single noisy images," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019.

[13] E. Kang, W. Chang, J. Yoo, and J. C. Ye, "Deep convolutional framelet denoising for low-dose CT via wavelet residual network," *IEEE Transactions on Medical Imaging*, vol. 37, no. 6, pp. 1358–1369, 2018.

[14] L. A. Feldkamp, L. C. Davis, and J. W. Kress, "Practical cone-beam algorithm," *Journal of the Optical Society of America A*, vol. 1, no. 6, pp. 612–619, 1984.

[15] J.-B. Thibault, K. D. Sauer, C. A. Bouman, and J. Hsieh, "A three-dimensional statistical approach to improved image quality for multislice helical CT," *Medical Physics*, vol. 34, no. 11, pp. 4526–4544, 2007.

Appendix

The project source code is available on the GitHub repository: https://github.com/Hermit-1710/SeniorThesis

Author’s Biography

1. Personal Information

Name: Yuxing Wu

Student ID: 3210110961

Major: Electrical Engineering

2. Education

2021.09–2026.06 ZJU-UIUC Institute, Undergraduate Program.

3. Research Experience

Senior thesis project on Transformer-based low-dose CT image denoising.

4. Awards, Publications, and Patents

2022 SRTP(Student Research Training Program) National Program, ZJU-UIUC Institute.